



Offline Policy Evaluation in Large Action Spaces via Outcome-Oriented Action Grouping

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ABSTRACT

Offline policy evaluation (OPE) aims to accurately estimate the performance of a hypothetical policy using only historical data, which has drawn increasing attention in a wide range of applications including recommender systems and personalized medicine. With the presence of rising granularity of consumer data, many industries started exploring larger action candidate spaces to support more precise personalized action. While inverse propensity score (IPS) is a standard OPE estimator, it suffers from more severe variance issues with increasing action spaces. To address this issue, we theoretically prove that the estimation variance can be reduced by merging actions into groups while the distinction among these action effects on the outcome can induce extra bias. Motivated by these, we propose a novel IPS estimator with outcome-oriented action Grouping (GroupIPS), which leverages a Lipschitz regularized network to measure the distance of action effects in the embedding space and merges nearest action neighbors. This strategy enables more robust estimation by achieving smaller variances while inducing minor additional bias. Empirically, extensive experiments on both synthetic and real world datasets demonstrate the effectiveness of our proposed method.

CCS CONCEPTS

• Computing methodologies → Causal reasoning and diagnostics.

KEYWORDS

Offline Policy Evaluation, Action Grouping, Embedding Space

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1 INTRODUCTION

In many interactive systems such as ad-placement, personalized healthcare and recommender systems [3, 30], it is ubiquitous to make personalized decisions following a contextual bandit setting. Policy plays a critical role that determines the displayed item or the recommended drug dosage (action) according to the characteristics of users (context), then observes the feedback (reward). The quality of a policy is evaluated by the utility which can be typically defined as the average response of users to the actions assigned by the policy. Precisely evaluating the utility of a new policy is paramount before deploying it in most real scenarios [9, 19]. The conventional approach for policy evaluation is online A/B tests [12]. A/B tests randomly assign actions to a subsample of the user populations, which can potentially be damaging to the user experience and even infeasible to implement especially in high-stakes industries including political strategy and personalized healthcare.

A recent strand of literature has extensively studied the evaluation of a policy based on logging data collected by a policy in the past [3, 13, 14, 24], which overcomes the two drawbacks above. The historical data consist of contexts, actions and the corresponding responses. Due to the action assignment mechanism of the behavior policy, the action is correlated with the contexts. This correlation induces distribution shifts in the historical data. Similarly, the target policy can also introduce potential distribution shifts into the observations when it is adopted on the population. In order to accurately evaluate the utility of the target policy based on the historical observations, it is necessary to eliminate the distribution discrepancy between the two shifted distributions induced by the target policy and the behavior policy.

The inverse propensity score (IPS) estimator [1, 8, 16, 24] is a common method for OPE, the main idea is to calculate sample weights based on the propensity score such that the distribution of contexts and actions under the target policy can be approximated by the reweighted observations. Although the IPS estimator enjoys the unbiased property when the propensity score is correctly specified, it suffers from the high variance issue when the importance weights are large. With the increasing amount of high granular user observations available, it is a desirable strategy for many industries to explore a larger candidate action space to support more precise personalized action assignment [2, 31]. For instance, promotional markdown is a common strategy for the e-commerce industry, the candidate spaces of possible discounts (such as 10% off, 15% off) or conditional discounts (such as 10% off for orders over 100\$) can be extremely large. Directly applying the IPS estimator under

such circumstances will result in large importance weights, which unavoidably lead to more severe high variance issue.

To alleviate this issue of exploring large action spaces, it is a sensible idea that many actions may have a similar impact on the average response of entire customer populations, which naturally motivates us to merge actions into groups. We theoretically analyze such action grouping method and characterize its bias and variance. We discover that under mild assumptions, it can potentially achieve substantial improvements on MSE through variance reduction. While such action grouping method can induce bias, our theoretical analysis indicates that the additional bias can be kept at a low level if the actions merged into the same group share similar effects on the outcome. In particular, the action effects can be roughly expressed as a mapping function from covariates to the outcome which is infinite-dimensional. Therefore, it comes with the challenge to investigate a feasible method that can effectively measure the difference of action effects using observational data.

Inspired by the motivation above, we propose a novel IPS estimator with outcome-oriented action Grouping (GroupIPS) algorithm, a practical approach to capture the action effects in the embedding space. To minimize the distinction of action effects within a group, we leverage Lipschitz regularization [4, 15] to ensure the consistency between the distance of action embeddings and their outcomes. Once we have learned the action embedding, our method can use any clustering approach to merge actions into groups based on their Euclidean distance in the embedding space. Furthermore, we conduct extensive experiments on both synthetic and real world dataset. Empirical results demonstrate that our method can outperform the existing baseline methods by effectively reducing the evaluation variances in various settings.

The main contribution of this paper are as following:

- We propose GroupIPS for OPE in large action spaces by merging similar actions into groups according to their effects on the outcome.
- Theoretical analysis reveals that GroupIPS can achieve smaller variance than other sample-reweighting methods while inducing minor additional bias, and setting the number of groups provides a controllable bias/variance trade-off.
- Extensive experimental results on both simulation and real world datasets present the effectiveness of GroupIPS compared with existing OPE estimators.

2 RELATED WORK

Existing offline policy evaluation methods can be categorized into several types, including direct methods, sample re-weighting based methods and the combination of them.

The direct methods learn a predictive model from the historical dataset to predict the counterfactual outcome and then use the predicted outcome to evaluate the utility of the target policy. Although many off-the-shelf machine learning algorithms can be directly applied for learning the counterfactual outcome, the direct methods may fail due to the distribution shifts in the historical dataset and the model misspecification.

The sample re-weighting based methods attempt to remove the distribution discrepancy of contexts and actions between the dataset generated by the behavior and the target policy. One of the most

typical methods is the inverse propensity score (IPS) estimator [13, 21, 24] for discrete action settings. The propensity score can be estimated with many machine learning models, such as logistic regression [16], and neural networks [29]. Under the correct model specification on the propensity score, it can be guaranteed that the IPS estimator is an unbiased estimator [7, 16]. However, this is not valid in many application scenarios. To overcome this obstacle, some literature [10] directly learn the sample weights by minimizing the distribution discrepancy between the weighted dataset and the dataset potentially generated by the target policy. To reduce the variance of IPS estimator, Swaminathan and Joachims [25] propose to normalize the original IPS weights and Kallus and Zhou [11] use kernel function to smooth the sample weights under the continuous setting. When the action is of more complex form than discrete variable, the propensity score estimation becomes more difficult, and Sondhi et al. [20] proposes to estimate the conditional density ratio of actions between the behavior policy and target policy by resolving a binary classification problem.

Recently, some literature attempt to combine the direct methods and sample re-weighting based methods [5, 6, 23, 26, 28] to incorporate the advantages of both methods. The doubly robust (DR) [5] method treats the results of the direct method as a baseline and subtract the baseline term from the outcome when re-weighting the samples. The unbiased property holds when either the propensity score model or the outcome prediction model is correct. The MAGIC estimator [26] is the weighted combination of IPS estimator and direct method. The SwitchDR estimator [28] switches the term of each sample between the predicted outcome and IPS re-weighted outcome according to the IPS sample weight. Based on the theoretical analysis on the bias and variance of the estimator, CAB [23] proposes to interpolate the direct method and IPS estimator in a sample-dependent manner and adaptively adjust the combined weight of predicted outcome and IPS re-weighted outcome for each sample according to the IPS sample weight.

Most of the previous works on offline policy evaluation concentrate on the binary action or relatively small action spaces. Saito and Joachims [18] propose MIPS estimator in large action spaces, which assumes prior additional action embeddings available and utilize the marginal distribution of action embeddings instead of raw actions for importance weights calculation, and theoretically prove that the degree of variance reduction depends on the quality of prior action embeddings. However, explicit action embedding is rarely available in real world scenarios, for instance, the embedding information for promotional discounts is difficult to be defined, and it is challenging to evaluate the quality of action embeddings, thus make the bias/variance tradeoff not controllable. Seldom literature have investigated to alleviate the high variance issues in the large action spaces via action spaces reduction without prior information.

3 PRELIMINARIES

In this section, we first provide the concepts and notations following the general contextual bandit setup. Then we list some classic methods for offline policy evaluation. Finally, we introduce our idea about merging actions with similar effects on outcome and analyze this impact on estimation bias and variance.

3.1 Problem Formulation

The dataset contains three elements, that is the context information $\mathbf{X} \in \mathcal{X} \subset \mathbb{R}^p$, which encodes the characteristics of each unit, action variable $\mathbf{A} \in \{1, 2, \dots, D\}$ where D is a large number and the corresponding outcome $\mathbf{r} \in \mathbb{R}$. Therefore, the historical dataset can be notated as $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{a}_i, \mathbf{r}_i)\}_{i=1,2,\dots,n}$, where n is the sample size. The process of collecting the dataset is:

- A unit with contexts $\mathbf{x}_i$ is sampled from the population distribution $\mathbf{x}_i \sim p(\mathbf{X})$.
- The action is assigned to the unit according to the behavior policy $\pi_0(\mathbf{A}|\mathbf{X})$, which takes context as input and returns the distribution of assigned actions. Formally, $\mathbf{a}_i \sim \pi_0(\mathbf{A}|\mathbf{x}_i)$.
- Given the context and assigned action, the outcome is revealed, that is $\mathbf{r}_i \sim p(\mathbf{r}|\mathbf{x}_i, \mathbf{a}_i)$.

With the collected dataset, we aim to estimate the utility $U(\pi)$ of target policy π , which demonstrate the performance of the policy given historical observations. Formally:

$$U(\pi) = \mathbb{E}_{\mathbf{X} \sim p(\mathbf{X}), \mathbf{A} \sim \pi(\mathbf{A}|\mathbf{X}), \mathbf{r} \sim p(\mathbf{r}|\mathbf{A}, \mathbf{X})} [\mathbf{r}]. \quad (1)$$

3.2 Existing OPE Estimators

Given the historical logged data $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{a}_i, \mathbf{r}_i)\}_{i=1,2,\dots,n}$ collected by a behavior policy π_0 , the existing estimators aim to estimate the utility mainly by two approaches, named "Model the World" and "Model the Bias" [23]. Below, we briefly introduce the two mainstreamed approaches for offline policy evaluation.

The direct method belongs to the "Model the World" approach. It learns a model $\hat{r}(\mathbf{X}, \mathbf{A})$ from the historical dataset to predict the expected outcome given the contexts and the action. The utility of target policy is estimated by:

$$\hat{U}_{\text{DM}}(\pi, \mathcal{D}) := \frac{1}{n} \sum_{i=1}^n \sum_{\mathbf{a}_i \in \mathcal{A}} \pi(\mathbf{a}_i|\mathbf{x}_i) \hat{r}(\mathbf{x}_i, \mathbf{a}_i). \quad (2)$$

The precise estimation of direct method relies on the correct model specification of outcome model, which is not the case in many scenarios. And the distribution shifts in the historical dataset can also exacerbate the performance drop of the model on the dataset potentially generated by the target policy.

Alternatively, sample re-weighting based method belongs to the "Model the Bias" approach. It re-weights the dataset to remove the distribution discrepancy between the datasets generated by the behavior policy and the target policy instead of imputing the counterfactual outcome. Inverse Propensity Score (IPS) estimator is one of the standard methods belongs to the "Model the Bias" approach:

$$\hat{U}_{\text{IPS}}(\pi, \mathcal{D}) := \frac{1}{n} \sum_{i=1}^n \frac{\pi(\mathbf{a}_i|\mathbf{x}_i)}{\pi_0(\mathbf{a}_i|\mathbf{x}_i)} \mathbf{r}_i. \quad (3)$$

Although it enjoys the unbiasedness property when the propensity score is correct, it usually suffers from high variances. The high variance problem would be more severe under large action spaces if IPS estimator is directly applied. In this work, we follow the "Model the Bias" approach and attempt to reduce the variance of the estimator and achieve more reliable policy evaluation for large action spaces.

3.3 Analysis on Action Grouping

Under the scenario with large action spaces, IPS estimator may suffer from variance inflation problem, which makes the estimation results unreliable.

To resolve this problem, we argue that we can merge actions into groups $\mathbf{g} \in \{1, 2, \dots, K\}$ and then regard the groups as substitute actions for IPS estimator. Theoretical analysis reveals that this strategy can reduce the variance while it may induce additional bias into the estimation. Fortunately, if the merged actions in the same group have a similar effect on the outcome, the additional bias can be controlled at a low level.

Definition 3.1. For each action $\mathbf{a} \in \{1, 2, \dots, D\}$, we define a corresponding variable $\mathbf{g} \in \{1, 2, \dots, K\}$ denoting the group that the action are merged into. Therefore, we can define the conditional distribution on the group variable given action as following:

$$p(\mathbf{g}|\mathbf{a}) = \begin{cases} 1 & \text{if } \mathbf{g} \text{ represents the group of } \mathbf{a} \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

The joint distribution on contexts, actions, groups and outcomes can also be defined as $p(\mathbf{x}, \mathbf{a}, \mathbf{r}, \mathbf{g}) = p(\mathbf{x}, \mathbf{a}, \mathbf{r})p(\mathbf{g}|\mathbf{a})$, where $p(\mathbf{x}, \mathbf{a}, \mathbf{r})$ is a data distribution generated by some policies (e.g. behavior policy or target policy).

Definition 3.2. Regarding the group variable as the substitute of the actions, we can obtain the action Grouping IPS estimator:

$$\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D}) = \frac{1}{n} \sum_{i=1}^n \frac{\pi(\mathbf{g}_i|\mathbf{x}_i)}{\pi_0(\mathbf{g}_i|\mathbf{x}_i)} \cdot \mathbf{r}_i, \quad (5)$$

where $\mathbf{g}_i$ represents the group of $\mathbf{a}_i$, $\pi(\mathbf{g}_i|\mathbf{x}_i) = \sum_{\mathbf{a}} \pi(\mathbf{a}|\mathbf{x}_i)p(\mathbf{g}_i|\mathbf{a})$ and $\pi_0(\mathbf{g}_i|\mathbf{x}_i)$ is similarly defined.

ASSUMPTION 1. We assume the outcome generation is of the form $\mathbf{r} = u(\mathbf{x}, \mathbf{a}) + \epsilon$, where $|u(\mathbf{x}, \mathbf{a})| \leq M_u$ and ϵ is a noise term with zero mean. We also assume the actions merged into one group have a similar effect on the outcome. It means that for arbitrary action pair $\mathbf{a}^1, \mathbf{a}^2$ satisfying $p(\mathbf{g}|\mathbf{a}^1) = p(\mathbf{g}|\mathbf{a}^2)$, we have

$$|\mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{a}^1] - \mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{a}^2]| \leq \delta, \forall \mathbf{x}, \quad (6)$$

where δ is a constant less than M_u . Furthermore, we constrain the discrepancy of two policies by assuming that there exists constants M_g and M_a satisfying for arbitrary $\mathbf{x}, \mathbf{g}, \mathbf{a}$:

$$\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} \leq M_g, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \leq M_a. \quad (7)$$

Assuming the statements above, we can have the following analysis on the bias and variance of the estimation.

PROPOSITION 3.3. If the Assumption 1 holds, we have the following result on the bias of the estimator $\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D})$,

$$\begin{aligned} & |\text{Bias}(\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D}))| \\ & \leq \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\sqrt{\text{Var}_{\mathbf{a} \sim \pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right]} \right] \cdot \frac{\delta}{2}. \end{aligned} \quad (8)$$

PROPOSITION 3.4. *If the Assumption 1 holds, we have the following results on the comparison between $\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D})$ and $\hat{U}_{\text{IPS}}(\pi, \mathcal{D})$:*

$$\begin{aligned} & n [\text{Var}(\hat{U}_{\text{IPS}}(\pi, \mathcal{D})) - \text{Var}(\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D}))] \\ & \geq \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\mathbb{E}[\mathbf{r}^2|\mathbf{x}, \mathbf{g}] \text{Var} \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right] \right] \\ & - (2M_a^2 M_u + 2M_u M_a + M_u M_g^2 M_a) \delta / 4. \end{aligned} \quad (9)$$

From the results in Proposition 3.3 and Proposition 3.4, if the actions merged into the same group have the similar effect on outcome (i.e. smaller δ), the $\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D})$ will reduce the variance based on $\hat{U}_{\text{IPS}}(\pi, \mathcal{D})$ while introducing minor additional bias. The detailed proof of the propositions can be found in appendix A.

The variance reduction of GroupIPS can be determined in two aspects, the first term is related to the variance of sample weights on the raw action spaces, which implies that GroupIPS can bring substantial variance reduction over IPS through merging actions with large variances on sample weights into groups. The second terms can be minor if the distinction of action effect within a group can be controlled at a low level.

Inspired by the conclusions above, we attempt to depict the distance of function $u(\cdot, \mathbf{a})$ between different actions and merge the actions with small distance into groups. Since the space of function $u(\cdot, \mathbf{a})$ is of infinite-dimension, it is intractable to compute the distance in the raw function space. Therefore, we propose to transform the outcome function $u(\cdot, \mathbf{a})$ into a finite-dimensional embedding and compute the distance in the projected space.

To realize this strategy, we first give the following assumption.

ASSUMPTION 2. *There exist a transformation function $\phi : \{D\} \rightarrow \mathbb{R}^{D_\phi}$ such that $\forall \mathbf{x}, |u(\mathbf{x}, \mathbf{a}^1) - u(\mathbf{x}, \mathbf{a}^2)| \leq \zeta \cdot \|\phi(\mathbf{a}^1) - \phi(\mathbf{a}^2)\|_2 + \eta$*

If we find this transformation ϕ satisfying the assumption 2 with a small value η , then we can cluster the action embeddings $\phi(\mathbf{a})$ based on the Euclidean distance so that the maximum distance of embeddings in one cluster can be controlled to a low value γ . Regarding the clusters as the action groups, the distance of outcome function $\delta \leq \zeta \cdot \gamma + \eta$ is also small.

To seek such transformation, we start from the following proposition:

PROPOSITION 3.5. *If we can find a good predictive regressor $f : \mathbb{R}^p \times \mathbb{R}^{D_\phi} \rightarrow \mathbb{R}$ and transformation function ϕ with Lipschitz constraints on action such that $|u(\mathbf{x}, \mathbf{a}) - f(\mathbf{x}, \phi(\mathbf{a}))| \leq \mu, \forall \mathbf{x}, \mathbf{a}$ and $|f(\mathbf{x}, \phi(\mathbf{a}^1)) - f(\mathbf{x}, \phi(\mathbf{a}^2))| \leq c \cdot \|\phi(\mathbf{a}^1) - \phi(\mathbf{a}^2)\|_2, \forall \mathbf{x}, \mathbf{a}^1, \mathbf{a}^2$. Then the function ϕ satisfies the Assumption 2 with $\zeta = c, \eta = 2\mu$.*

Inspired by the Proposition 3.5, we propose the skeleton of our algorithm which is learning the transformation ϕ and a Lipschitz regularized regressor u with small predictive error (i.e. small μ). In the next section, we instantiate the idea and propose our practical algorithms.

4 ALGORITHM

In this section, we present the detailed description of our algorithm. Since we group the actions based on the outcome functions, we name our algorithm as outcome-oriented action Grouping IPS

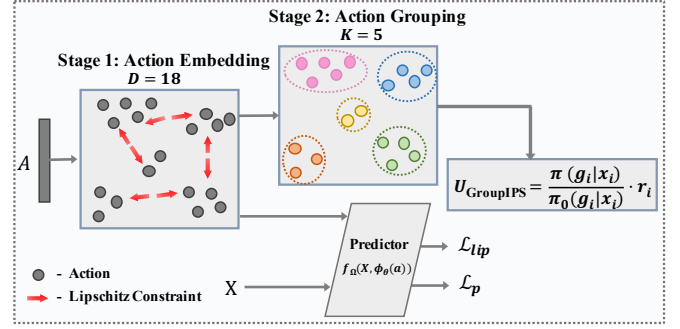


Figure 1: The framework of GroupIPS, the first stage is to construct the outcome-oriented action embedding by jointly optimizing the Lipschitz loss $\mathcal{L}_{lip}$ and prediction loss $\mathcal{L}_p$. The second stage merges action a_i into group g_i using clustering method and perform GroupIPS.

(GroupIPS) method. The method consists of outcome-oriented action embedding learning and action grouping. The overall framework of GroupIPS is shown in figure 1.

4.1 Outcome-oriented Action Embedding

We use neural networks to construct the action embedding, to ensure that the distance of action embeddings can reflect the distance of their effects on the outcome. To achieve this, we construct the model acting as f in Proposition 3.5 which predicts the outcome based on the action embeddings with Lipschitz regularization.

4.1.1 *Action embedding.* We apply a neural network ϕ parameterized by θ with fully-connected layers as the action embedding:

$$\phi_\theta : [D] \rightarrow \mathbb{R}^{D_\phi}, \quad (10)$$

where $[D] = \{1, 2, \dots, D\}$ and it takes the raw action variable as input and output the action embedding that encodes the distance of action effects on outcomes.

4.1.2 *Outcome-oriented predictive module.* To guarantee that the learned embedding $\phi_\theta(a_i)$ reflects the action effects on the outcome and the corresponding distance between each other, we concatenate the representation $\phi_\theta(a_i)$ with the context $\mathbf{X}$ and put the concatenation into a predictive networks $f_\Omega : \mathbb{R}^p \times \mathbb{R}^{D_\phi} \rightarrow \mathbb{R}$ to predict the outcome. The loss for encouraging the predictive information is defined as:

$$\mathcal{L}_p = \frac{1}{n} \sum_{i=1}^n (f_\Omega(\mathbf{x}_i, \phi_\theta(\mathbf{a}_i)) - y_i)^2, \quad (11)$$

4.1.3 *Lipschitz Regularization.* In order to ensure the consistency that the distance in the embedding space can accurately reflect the relationship of action effects, we leverage Lipschitz regularization such that f_Ω is Lipschitz continuous if there exists a constant $c \geq 0$ such that:

$$|f_\Omega(z_i) - f_\Omega(z_j)| \leq c \|z_i - z_j\|_2, \quad (12)$$

where $z_i = (\mathbf{x}_i, \phi_\theta(\mathbf{a}_i))$ is the concatenation of the context and the action embedding, and intuitively, the change of outcome is

bounded by constant c for smoothness. It is obvious that if the neural networks f_Ω is c -Lipschitz on the z , it is also c -Lipschitz on the action embedding $\phi_\theta(a)$.

We leverage Lipschitz Regularization [15] which solely depends on the weight matrices W_i of each network layer with estimated per-layer Lipschitz upper bound c_i , the loss for regularizing the differences between action effects and outcomes can be written as:

$$\mathcal{L}_{Lip} = \sum_{i=1}^l \text{softplus}(c_i), \quad (13)$$

Where $\text{softplus}(c_i) = \ln(1 + e^{c_i})$ is a reparameterization strategy to avoid invalid negative estimation on Lipschitz constant c_i and l is the number of network layers.

Summarily, the final loss function for outcome-oriented action embedding is:

$$\mathcal{L} = \mathcal{L}_p + \alpha \mathcal{L}_{Lip}, \quad (14)$$

where α is the hyper-parameter to control the strength of aligning the difference between the action embedding and the outcome. The tuning of α is stated to be minimal in [15], we set it to be 10^{-3} in all experiments. By simultaneously optimizing $\mathcal{L}$ and $\mathcal{L}_{Lip}$, we can obtain a predictive model and the embedding function satisfying the requirements in Proposition 3.5.

4.2 Action Grouping

Once we have learned the action embedding space, we can use it for action grouping. Intuitively, our goal is to minimize the discrepancies of actions within the same group given the hyperparameter number of groups K . As we theoretically analyze the bias and variance of action grouping in section 3.3, with given K groups, we can achieve minor additional bias by constraining the distance of each action embedding $\phi_\theta(a_i) \in \mathbf{g}_i$ to the centroid of their group by Euclidean distance:

$$\|\phi_\theta(a_i) - \frac{1}{|S(\mathbf{g}_i)|} \sum_{j \in S(\mathbf{g}_i)} \phi_\theta(a_j)\|_2. \quad (15)$$

Our model can use any clustering mapping $h : \mathbb{R}^{D_\phi} \rightarrow [K]$ such that $h(\phi_\theta(a_i)) = \mathbf{g}_i$ that achieves the above objective and $S(\mathbf{g}_i) = \{j | h(\phi_\theta(a_j)) = \mathbf{g}_i\}$ is the set of samples in group $\mathbf{g}_i$. After applying the action grouping method to the original dataset $\mathcal{D}$, we can obtain the transformed observations $\mathcal{D}' = \{(\mathbf{x}_i, \mathbf{g}_i, r_i)\}_{i=1,2,\dots,n}$, then we can use $\mathcal{D}'$ to estimate the utility of GroupIPS using the group variable $\mathbf{g}$ as the substitute of action $\mathbf{a}$ following equation 5.

Summarily, the pseudo-code in Algorithm 1 depicts the detailed description of our GroupIPS method.

5 EXPERIMENTS

In this section, we empirically evaluate the effectiveness of our proposed GroupIPS estimator on both synthetic and large-scale real world datasets.

5.1 Baseline Estimator

We compare GroupIPS estimator with IPS [8], SNIPS [25], DM trained with Random Forest, Doubly Robust (DR) [5] and SwithchDR [28]. For completeness, we also include Lipschitz MLP (LipMLP)

Algorithm 1 Outcome-oriented action Grouping IPS (GroupIPS)

Input: Historical dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{a}_i, r_i)\}_{i=1,2,\dots,n}$ generated by the behavior policy π_0 , target policy π , learning rate λ , hyper-parameter α .

Output: The estimated utility of target policy $\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D})$.

- 1: Initialize the parameter θ of outcome-oriented action embedding component.
 - 2: **for** Batch $B = \{(\mathbf{x}_i, \mathbf{a}_i, y_i)\}_{i=1}^{|B|}$ in each iteration **do**
 - 3: $\mathcal{L}_p \leftarrow \frac{1}{|B|} \sum_{i=1}^{|B|} (f_\Omega(\mathbf{x}_i, \phi_\theta(\mathbf{a}_i)) - y_i)^2$
 - 4: $\mathcal{L}_{Lip} \leftarrow \frac{1}{|B|} \sum_{i=1}^{|B|} \prod_{l=1}^l \ln(1 + e^{c_l})$
 - 5: $\mathcal{L} \leftarrow \mathcal{L}_p + \alpha \mathcal{L}_{Lip}$
 - 6: Update $\theta \leftarrow \theta - \lambda \cdot \frac{\partial \mathcal{L}}{\partial \theta}$.
 - 7: **end for**
 - 8: Perform action grouping clustering $\mathbf{g} = h(\phi_\theta(\mathbf{a}))$
 - 9: Transform $\mathcal{D}$ into $\mathcal{D}' = \{(\mathbf{x}_i, \mathbf{g}_i, r_i)\}_{i=1,2,\dots,n}$
 - 10: $\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D}') = \frac{1}{n} \sum_{i=1}^n \frac{\pi(\mathbf{g}_i | \mathbf{x}_i)}{\pi_0(\mathbf{g}_i | \mathbf{x}_i)} \cdot r_i$
 - 11: **return** $\hat{U}_{\text{GroupIPS}}(\pi, \mathcal{D})$
-

[15] which is the network that we utilized to learn the action embedding space to regress the historical data, and GroupIPS (Oracle) that leverages the known action grouping mechanism to reweight the units.

5.2 Synthetic Experiment

5.2.1 Setup. We sample a range of synthetic datasets under various settings to examine the effectiveness and robustness of GroupIPS estimator. For each sample, we generate p contexts $\mathbf{X} = (x_{1,p}, x_{2,p}, \dots, x_{p,p})$ independently with Gaussian distribution:

$$x_{i,p} \stackrel{iid}{\sim} \mathcal{N}(0, 1), 1 \leq i \leq p. \quad (16)$$

The reward of each observation is generated as following:

$$r(\mathbf{x}_i, a) = -\beta_1^T \mathbf{x}_i \cdot e^{\beta_2^T \mathbf{x}_i} + |\beta_3^T \mathbf{x}_i + h(a)|^{0.8} + \epsilon_i, \quad (17)$$

Where $\beta_1, \beta_2, \beta_3 \in \mathbb{R}^{p \times 1}$ are set of parameter vectors, these parameters are sampled from vector $\mathbf{o}_i$ from $\text{Unif}(0, 1)$, and then setting $\beta_i = \mathbf{o}_i / \|\mathbf{o}_i\|$, where $\|\cdot\|$ is Euclidean norm. The effect of each action $\mathbf{a} \in \{1, 2, \dots, D\}$ is assigned by $h(\mathbf{a}_i) = N(\mu, \sigma)$, we set $\mu = \lfloor \frac{i}{K^*} \rfloor * \frac{3}{M}$, where K is the number of actions that share the similar effect, when $K^* = 1$, each action has distinct effects on the outcome. We set K^* to be 5, and M denotes the number of action groups. The variance of action effect inside a group is controlled by $\sigma = \frac{1}{D}$.

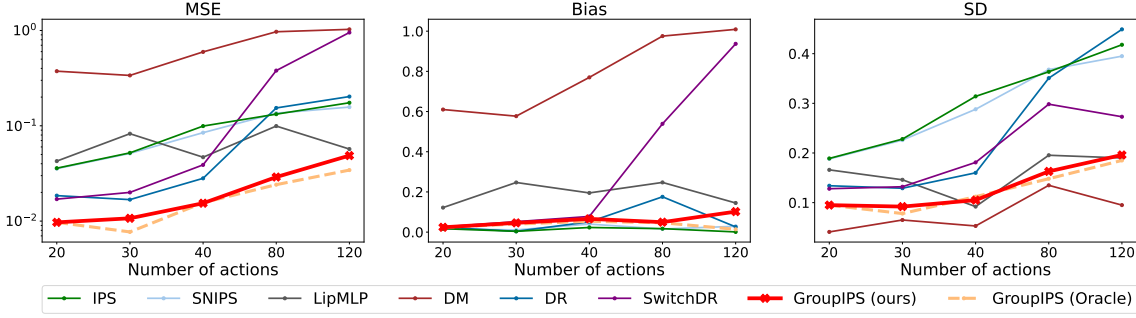
We generate behavior policy π_0 by applying softmax function to the reward similarly to [18]:

$$\pi_0(a_i | \mathbf{x}_i) = \frac{\beta \cdot e^{r(\mathbf{x}_i, a_i)}}{\sum_{a_j \in \mathbf{A}} \beta \cdot e^{r(\mathbf{x}_i, a_j)}}, \quad (18)$$

where β is a parameter that controls the optimality of the behavior policy. With the access of groundtruth rewards, as the value of β increases, we obtain a nearly optimal and deterministic behavior policy.

Table 1: The experimental results on the synthetic dataset. The smaller metric means better performance, and the best performance of MSE is marked bold.

Fix the number of actions $D = 100$, sample size $n = 10000$, varying the selection bias strength β of behavior policy												
β	$\beta = -2$			$\beta = -1$			$\beta = -0.5$			$\beta = 0$		
Methods	MSE	Bias	SD	MSE	Bias	SD	MSE	Bias	SD	MSE	Bias	SD
$\hat{U}_{\text{IPS}}$	0.6724	0.133	0.809	0.4300	0.135	0.642	0.3399	0.018	0.583	0.3585	0.002	0.599
$\hat{U}_{\text{SNIPS}}$	1.4648	0.115	1.205	0.3580	0.104	0.589	0.3372	0.001	0.581	0.3606	0.005	0.600
$\hat{U}_{\text{DM}}$	1.5511	0.303	1.208	0.9064	0.946	0.106	0.6463	0.802	0.052	0.3870	0.621	0.038
$\hat{U}_{\text{DR}}$	0.6476	0.248	0.766	0.3107	0.024	0.557	0.3517	0.008	0.593	0.3505	0.020	0.592
$\hat{U}_{\text{SwitchDR}}$	0.6482	0.343	0.728	0.7191	0.609	0.590	0.3759	0.224	0.571	0.3505	0.020	0.592
$\hat{U}_{\text{LipMLP}}$	0.7165	0.685	0.498	0.1793	0.020	0.423	0.5561	0.536	0.518	0.3114	0.369	0.418
$\hat{U}_{\text{GroupIPS}}$	0.3795	0.363	0.498	0.1369	0.117	0.351	0.2221	0.131	0.453	0.1972	0.028	0.443

**Figure 2: Performance of estimators under different number of action D .**

We set the target policy to synthesize a well-trained decision model:

$$\pi(a_i|x_i) = \gamma \cdot \mathbb{1}\{a_i = \arg \max_{a' \in A} [r(x_i, a')]\} + \frac{1-\gamma}{D}, \quad (19)$$

where $\gamma \in [0, 1]$ is the parameter that controls the optimality of the target policy, we set γ to be 0.8, which indicates that the action with the optimal outcome has a probability of 0.8 to be assigned, and the selection of other actions are with equal probabilities of $\frac{1-0.8}{D}$.

Under various settings of varying selection bias strength β , dimension of action D and sample size N , the ground truth utility can be computed as:

$$U(\pi) = \frac{1}{n} \sum_{i=1}^n \sum_{a_j \in A} r(x_i, a_j) \pi(a_j|x_i). \quad (20)$$

In offline policy evaluation, bias is a metric that evaluates the ability of an estimator to handle distribution shifts. However, since in most real scenarios, it is infeasible to repeatedly sample the actions from the behavior policy and conduct offline policy evaluation several times, we argue that MSE is a more crucial and representative metric. In this way, a comprehensive estimator can be validated simultaneously from the perspective of stability (SD) and handling distribution shifts (Bias).

5.2.2 Results. We conduct repeated experiments for 20 times under each setting and record the *Bias*, *SD*, *MSE* of each estimator.

Can GroupIPS perform well under different selection bias strength β of behavior policy? As results demonstrated in table

1, we fix the number of actions $D = 100$, sample size $N = 10,000$, and we vary the selection bias strengths of the behavior policy β in $[-2, -1, -0.5, 0]$, the lower the value of β means less discrepancy between the behavior and target policies. We can observe that for both IPS and SNIPS estimators, the variance becomes larger with the increasing discrepancy between behavior and target policies, which is due to large sample weights.

GroupIPS outperformed other estimators across all different behavior policies. Compared with IPS, GroupIPS trades minor additional bias with large variance reduction, thus achieving consistently better evaluation MSE. The performance advantage of GroupIPS over IPS becomes larger with increasing policies discrepancy, which further verify that GroupIPS can alleviate the variance issue of IPS caused by large sample weights. DM and LipMLP suffer from high bias due to model misspecification. DR performs relatively better than IPS as expected by combining both IPS and DM methods for estimation.

How effective is GroupIPS when we have a larger action spaces D ? The goal of this paper is primarily alleviating the high variance issue when extending sample re-weighting methods to a larger action space, to verify the performance of our proposed GroupIPS with other estimators in increasing action spaces, we fix the number of samples $N = 10,000$, $\beta = -1$, and vary the number of actions D in $[20, 30, 40, 80, 120]$. Figure 2 shows that the performance of IPS and SNIPS have become worse than LipMLP when D is larger than 30 despite achieving lower bias, the MSE of IPS and SNIPS started to rise rapidly due to the high estimation

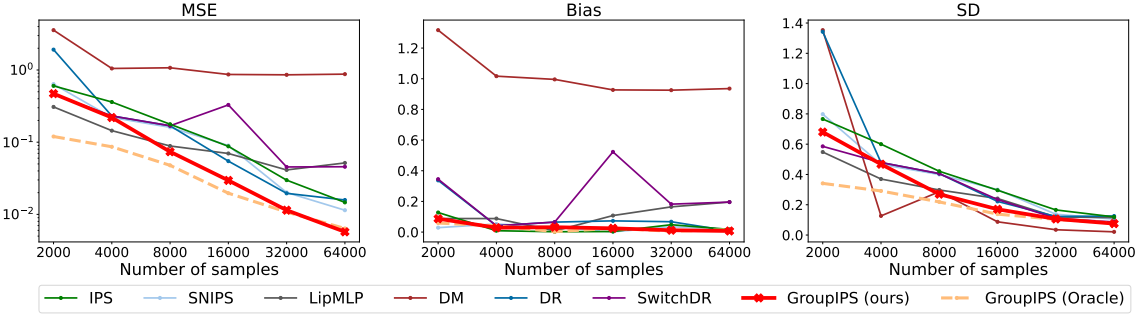


Figure 3: Performance of estimators under different number of sample size N .

variances. GroupIPS (Oracle) can still outperform LipMLP and DM when $D = 120$, which illustrates that sample re-weighting method can be extended to larger action spaces with much lower bias if we can effectively merge actions into groups. GroupIPS consistently demonstrates substantial improvements over IPS while even achieving similar results to GroupIPS (Oracle) when the action space is smaller than 120, indicating the accurate measurement of action effects in the learned embedding space. With the increasing number of actions, the estimation variances of LipMLP, DM are small but remaining highly biased.

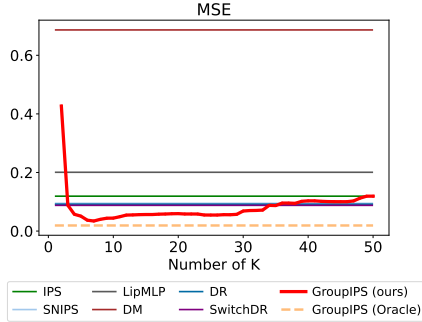


Figure 4: Performance of GroupIPS under different hyper-parameter K fixing the number of action $D = 50$.

How does sample size N affect the performance of GroupIPS?

We further evaluate the performance of GroupIPS with increasing sample size N in $[2,000, 4,000, 8,000, 16,000, 32,000, 64,000]$, we fix the number of actions $D = 100$ and $\beta = -1$. As results shown in figure 3, the MSE improvement of GroupIPS over IPS becomes increasingly larger with more samples N . GroupIPS can obtain a action embedding space to accurately measure the action effects and achieves similar results with GroupIPS (Oracle) from $N = 8,000$ which outperforms all other estimators.

With more samples available, the evaluation MSE of DM and LipMLP estimator do not decrease, we can observe that the bias remains high due to simply ignoring the shifts between the behavior and target policies. The MSE of IPS, SNIPS and DR decrease and outperform LipMLP and SwitchDR when the sample size is over 16,000 with lowered variances. We also find that with reduced

action spaces, GroupIPS can achieve the same evaluation MSE with significantly less samples.

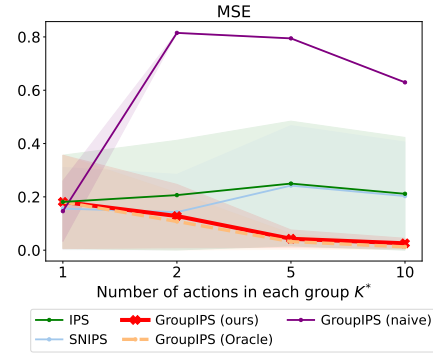


Figure 5: MSE comparison with variance shaded of different number of actions in each group K^* .

5.2.3 Ablation Study.

How robust is GroupIPS when the number of actions in each group is unknown? We vary the hyper-parameter number of groups K for GroupIPS to verify the robustness to the group selection. We set the number of actions $D = 50$, and the sample size $N = 10,000$, $\beta = -1$, the number of actions in each group $K^* = 5$, then we vary the number of groups K from $[1, 50]$. The results on MSE is reported in Figure 5. GroupIPS performs quite poorly when we have $K = [1, 2, 3, 4]$ as expected due to large bias by simply ignoring the distinction of action effects in the same group. When we choose a reasonable group number K from $[5, 50]$, GroupIPS improves substantially over IPS and we can observe a clear bias/variance tradeoff as we change the value of K . The performance of GroupIPS remains stable and it illustrates a wide range of the candidate values K that can improve the evaluation performance. Note that GroupIPS is equivalent to IPS when $K = D$ with unmodified action spaces. We include the bias and variance results and the additional experiment for $D = 100$ in appendix section A.

Sensitivity of GroupIPS to the number of actions in each cluster K^* . In real world scenarios, the underlying action effects

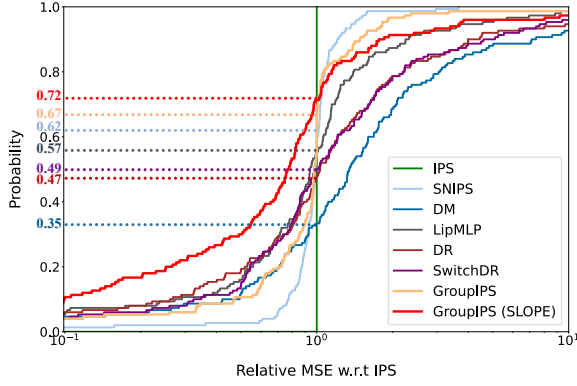


Figure 6: Relative MSE of estimators to IPS.

are typically unknown, inspired by this, we vary the number of actions in each group K^* to verify GroupIPS’s robustness. We fix $D = 50$, $N = 10,000$, and we include GroupIPS (naive) that naively group actions by their action indexes and ignore the action effects on the outcome. As demonstrated in figure 5, note that GroupIPS is equivalent to IPS when we set $K = 1$ in this setting. GroupIPS outperforms IPS across all settings, the substantial performance gains are from effective variance reduction as demonstrated by the variances shaded in figure 5. The performance gap has become increasingly large when we increase the value of K^* that results in more actions sharing similar effects on the outcome. GroupIPS (naive) fails in all settings by simply ignoring the information of the outcome with high bias.

5.3 Real World Experiment

5.3.1 Dataset. To further demonstrate the effectiveness of GroupIPS, we evaluate its performance on Open Bandit Dataset (OBD) [17], a large-scale real world bandit dataset. OBD dataset conducted A/B tests under either uniform random policy or Bernoulli Thompson Sampling on a fashion e-commerce platform. We use Men’s campaigns provided by OBP that contain 102 actions ($D = 102$) and the utilities of two policies are the average click-through rate (CTR) of customers under 7-day A/B tests, and we set random and Bernoulli Thompson Sampling policies as our behavior and target policies respectively.

Data-driven group selection method. We have demonstrated the stability of GroupIPS’s performance with varying group K in 5.2.3. Furthermore, we introduce a statistic data-driven method SLOPE [22, 27] to provide an automatic K selection which can compete with oracle selection for the bias/variance tradeoff. To demonstrate its performance in the real world dataset, we include GroupIPS (SLOPE) that uses SLOPE in the experiment.

5.3.2 Results. Robustness of GroupIPS in real world data. We bootstrapped 150 times from the logged dataset, and each time we sample 10,000 observations. To better visually demonstrate the performance of each estimator, we record the MSE normalized by that of IPS. Figure 6 draws the cumulative distribution function (CDF), which we can observe that GroupIPS outperforms IPS for

Methods	$N = 10,000$	$N = 20,000$	$N = 50,000$
$\hat{U}_{\text{IPS}}$	7.250%	3.917%	3.211%
$\hat{U}_{\text{SNIPS}}$	7.155%	3.911%	3.217%
$\hat{U}_{\text{DM}}$	8.867%	4.373%	3.334%
$\hat{U}_{\text{DR}}$	8.543%	4.110%	3.785%
$\hat{U}_{\text{SwitchDR}}$	8.084%	4.328%	3.571%
$\hat{U}_{\text{LipMLP}}$	7.130%	4.391%	3.171%
$\hat{U}_{\text{GroupIPS}}$	<u>7.030%</u>	<u>3.571%</u>	2.929%
$\hat{U}_{\text{GroupIPS(SLOPE)}}$	6.758%	3.071%	<u>2.966%</u>

Table 2: Relative MSE of varying bootstrapped sample size, the best and the second best performance is marked as bolded and underlined respectively.

about 67% bootstrapped evaluations, where by incorporating the data-driven group K selection technique SLOPE, we can achieve even better performance at 72%. DM fails with only 35% of bootstrapped evaluation better than IPS. SwitchDR and DR performed slightly worse than IPS due to the poor result of DM, while SNIPS and LipMLP demonstrate relatively better results that outperformed IPS for 62% and 57% of 150 bootstrapped evaluations respectively.

How does GroupIPS perform with varying observations?

In addition, in table 2, we vary the number of observations bootstrapped at each time, and record the relative MSE that is the MSE of each estimator divided by the groundtruth utility of the target policy. The results demonstrate the robustness of GroupIPS in real world application, GroupIPS and GroupIPS (SLOPE) outperform the other estimators across all settings, which further prove that SLOPE can provide automatic K selection for GroupIPS with reliable evaluation.

Sample efficiency of GroupIPS. As demonstrated in table 2, GroupIPS demonstrates its sample efficiency by reducing the large action spaces using outcome-oriented action grouping. GroupIPS achieves comparable evaluation performance under $N = 20,000$ with IPS under $N = 50,000$, by leveraging SLOPE for automatic K selection, GroupIPS (SLOPE) even outperforms IPS using only 40% of data, we can observe a similar phenomenon in synthetic studies that GroupIPS is sample efficient compared with IPS, with increasing action spaces, GroupIPS is a more desirable estimator.

6 CONCLUSIONS

In this paper, our aim is to perform offline policy evaluation in large action spaces. We state that IPS on the raw action spaces can result in high variance issue. To address this issue, we propose to merge actions into groups that share similar effects on the outcome. We theoretically characterize the bias, variance of the action grouping method and the possible MSE gains. Inspired by the above motivation, we propose GroupIPS that merges action into groups based on a Lipschitz regularized action embedding space and utilize the groups to calculate the sample weights following the standard IPS method. Extensive results on both synthetic and real world dataset demonstrate the effectiveness and robustness of GroupIPS. Moreover, GroupIPS provides a controllable bias/variance trade-off for OPE through the number of groups K .

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A PROOF

PROPOSITION A.1. (**Restated**) If the Assumption 1 holds, we have the following result on the bias of the estimator $\hat{U}_{GroupIPS}(\pi, \mathcal{D})$,

$$\begin{aligned} & |Bias(\hat{U}_{GroupIPS}(\pi, \mathcal{D}))| \\ & \leq \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\sqrt{Var_{\mathbf{a} \sim \pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right]} \right] \cdot \frac{\delta}{2} \quad (21) \end{aligned}$$

PROOF.

$$\begin{aligned} & |Bias(\hat{U}_{GroupIPS}(\pi, \mathcal{D}))| \\ & = |\mathbb{E}[\hat{U}_{GroupIPS}(\pi, \mathcal{D})] - \mathbb{E}[\hat{U}_{IPS}(\pi, \mathcal{D})]| \\ & = \left| \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi_0(\mathbf{g}|\mathbf{x}), \mathbf{a} \sim \pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \left[\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} \mathbf{r} \cdot \left(1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right) \middle| \mathbf{x}, \mathbf{g}, \mathbf{a} \right] \right| \\ & = \left| \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi_0(\mathbf{g}|\mathbf{x})} \left[\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} \mathbb{E} \left[\mathbf{r} \cdot \left(1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right) \middle| \mathbf{x}, \mathbf{g} \right] \right] \right| \\ & = \left| \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\mathbb{E} \left[\mathbf{r} \cdot \left(1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right) \middle| \mathbf{x}, \mathbf{g} \right] \right] \right| \end{aligned}$$

Since $\mathbb{E} \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right] = 1$, we have

$$\begin{aligned} & \left| \mathbb{E} \left[\mathbf{r} \cdot \left(1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right) \middle| \mathbf{x}, \mathbf{g} \right] \right| \\ & = \left| \mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{g}] \cdot \mathbb{E} \left[1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right] + Cov \left(\mathbf{r}, 1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right) \right| \\ & = \left| Cov \left(\mathbf{r}, 1 - \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right) \right| \\ & \leq \sqrt{Var(u(\mathbf{x}, \mathbf{a})) \cdot Var \left(\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right)} \\ & \leq \frac{\delta}{2} \cdot \sqrt{Var_{\mathbf{a} \sim \pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \right]} \end{aligned}$$

Therefore, we have the conclusion in the proposition. $\square$

PROPOSITION A.2. (**Restated**) If the Assumption 1 holds, we have the following results on the comparison between $\hat{U}_{GroupIPS}(\pi, \mathcal{D})$ and $\hat{U}_{IPS}(\pi, \mathcal{D})$:

$$\begin{aligned} & n [Var(\hat{U}_{IPS}(\pi, \mathcal{D})) - Var(\hat{U}_{GroupIPS}(\pi, \mathcal{D}))] \\ & \geq \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\mathbb{E}[\mathbf{r}^2|\mathbf{x}, \mathbf{g}] Var \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right] \right] \\ & - (2M_a^2 M_u + 2M_u M_a + M_u M_g^2 M_a) \delta / 4 \end{aligned}$$

PROOF. According to the law of total variance, we have

$$\begin{aligned} & n [Var(\hat{U}_{IPS}(\pi, \mathcal{D})) - Var(\hat{U}_{GroupIPS}(\pi, \mathcal{D}))] \\ & = Var_{\mathbf{x}, \mathbf{g} \sim \pi_0(\mathbf{g}|\mathbf{x})} \left[\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} Cov(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g}) \right] \\ & + Cov_{\mathbf{x}, \mathbf{g} \sim \pi_0(\mathbf{g}|\mathbf{x})} \left[\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} \mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{g}], Cov(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g}) \right] \\ & + \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\mathbb{E}[\mathbf{r}^2|\mathbf{x}, \mathbf{g}] Var \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right] \right] \\ & + Cov \left(\mathbf{r}^2, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})^2}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})^2} \middle| \mathbf{x}, \mathbf{g} \right) - Cov^2 \left(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right) \\ & - 2\mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{g}] Cov \left(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})^2}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right) \end{aligned}$$

Because for any random variable takes value in the interval with width e , the variance of it is less than $e^2/4$.

$$\begin{aligned} & Var_{\mathbf{x}, \mathbf{g} \sim \pi_0(\mathbf{g}|\mathbf{x})} \left[\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} Cov(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g}) \right] \geq 0 \\ & Cov_{\mathbf{x}, \mathbf{g} \sim \pi_0(\mathbf{g}|\mathbf{x})} \left[\frac{\pi(\mathbf{g}|\mathbf{x})}{\pi_0(\mathbf{g}|\mathbf{x})} \mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{g}], Cov(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g}) \right] \\ & \geq -M_u M_g^2 M_a \delta / 4 \\ & \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[Cov \left(\mathbf{r}^2, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})^2}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})^2} \middle| \mathbf{x}, \mathbf{g} \right) \right] \geq -M_a^2 (2M_u - \delta) \delta / 4 \\ & \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[Cov^2 \left(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right) \right] \geq -M_a^2 \delta^2 / 4 \\ & -2\mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\mathbb{E}[\mathbf{r}|\mathbf{x}, \mathbf{g}] Cov \left(\mathbf{r}, \frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})^2}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right) \right] \geq -M_u \delta M_a / 2 \end{aligned}$$

Therefore, we have

$$\begin{aligned} & n [Var(\hat{U}_{IPS}(\pi, \mathcal{D})) - Var(\hat{U}_{GroupIPS}(\pi, \mathcal{D}))] \\ & \geq \mathbb{E}_{\mathbf{x}, \mathbf{g} \sim \pi(\mathbf{g}|\mathbf{x})} \left[\mathbb{E}[\mathbf{r}^2|\mathbf{x}, \mathbf{g}] Var \left[\frac{\pi(\mathbf{a}|\mathbf{x}, \mathbf{g})}{\pi_0(\mathbf{a}|\mathbf{x}, \mathbf{g})} \middle| \mathbf{x}, \mathbf{g} \right] \right] \\ & - (2M_a^2 M_u + 2M_u M_a + M_u M_g^2 M_a) \delta / 4 \end{aligned}$$

$\square$

PROPOSITION A.3. (**Restated**) If we can find a good predictive regressor $f: \mathbb{R}^p \times \mathbb{R}^{D_\phi} \rightarrow \mathbb{R}$ and transformation function ϕ with lipschitz constraints on action such that $|u(\mathbf{x}, \mathbf{a}) - f(\mathbf{x}, \phi(\mathbf{a}))| \leq \mu, \forall \mathbf{x}, \mathbf{a}$ and $|f(\mathbf{x}, \phi(\mathbf{a}^1)) - f(\mathbf{x}, \phi(\mathbf{a}^2))| \leq c \cdot \|\phi(\mathbf{a}^1) - \phi(\mathbf{a}^2)\|_2, \forall \mathbf{x}, \mathbf{a}^1, \mathbf{a}^2$. Then the function ϕ satisfies the Assumption 2 with $\zeta = c, \eta = 2\mu$.

PROOF.

$$\begin{aligned} & |u(\mathbf{x}, \mathbf{a}^1) - u(\mathbf{x}, \mathbf{a}^2)| \leq |f(\mathbf{x}, \phi(\mathbf{a}^1)) - f(\mathbf{x}, \phi(\mathbf{a}^2))| \\ & + |f(\mathbf{x}, \phi(\mathbf{a}^1)) - u(\mathbf{x}, \mathbf{a}^1)| + |f(\mathbf{x}, \phi(\mathbf{a}^2)) - u(\mathbf{x}, \mathbf{a}^2)| \\ & \leq c \cdot \|\phi(\mathbf{a}^1) - \phi(\mathbf{a}^2)\|_2 + 2\mu \end{aligned}$$

$\square$

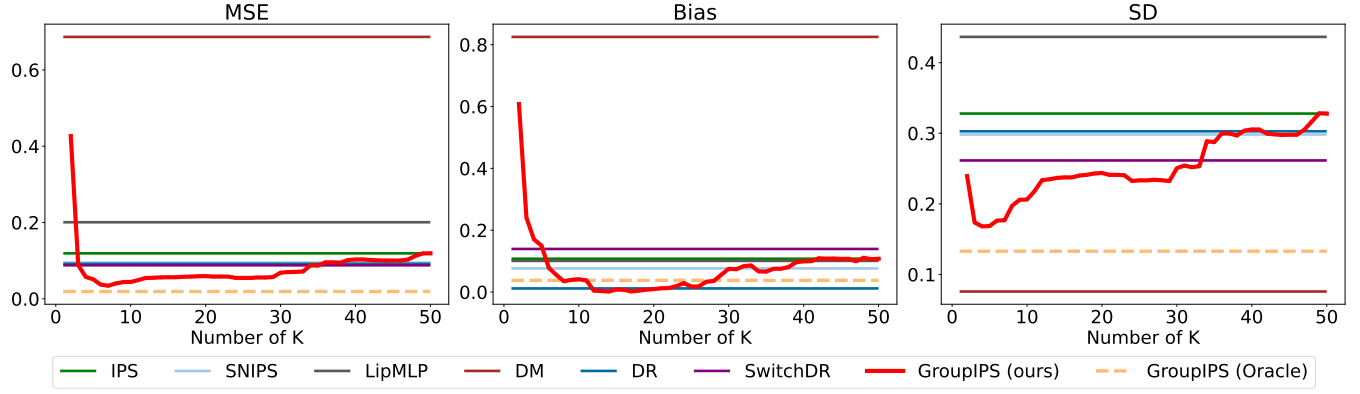


Figure 7: Fix the number of actions $D = 50$, performance of estimators under different number of clusters.

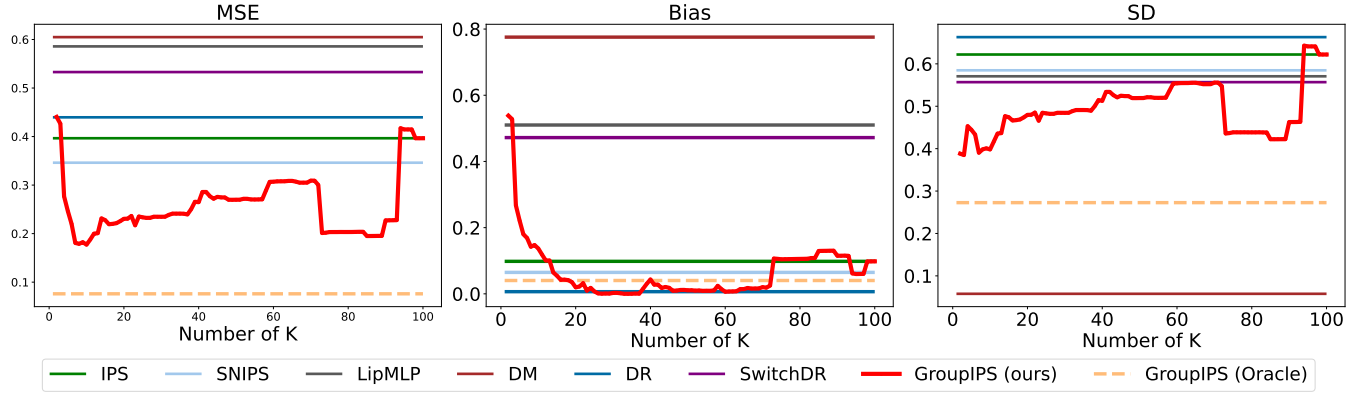


Figure 8: Fix the number of actions $D = 100$, performance of estimators under different number of clusters.

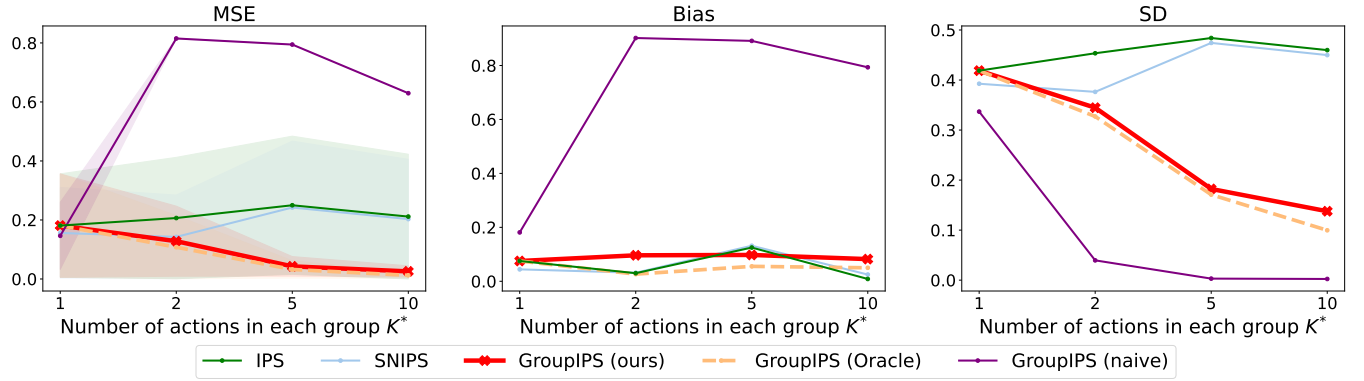


Figure 9: MSE comparison of estimators under different number of actions in each group K^* .

B ADDITIONAL EXPERIMENTS

In this section, we provide additional experimental results and the bias and standard deviation of each estimator. We provide the results of bias and variance for each estimator for 5.2.3 in figure 7 and figure 9, which demonstrate a similar phenomenon that GroupIPS can trade minor additional bias for large variance reduction, hence

achieving better MSE to provide more reliable evaluation results. We perform similar experiment to 5.2.3 and we set number of actions $D = 100$, the number of actions in each group $K^* = 20$, in figure 8, we can observe similar trend that the bias is large than IPS when we vary K from $[1, 10]$, but it achieves significant variance reductions, which results in MSE improvements over IPS.